

JORGE MOLINA

✉ contato@jorgemolina.dev

☎ +55 (12)98114-5523

📍 Brazil - Remote

🔗 jorgemolina.dev

🐙 joorgem

🌐 jorge-molinadavid

AI DATA ENGINEER • AI ENGINEER
SNOWFLAKE • PYTHON • SQL

SUMMARY

Senior AI Data Engineer specializing in scalable data platforms and production LLM systems using Python and SQL. Experienced in transforming unstructured data into structured, reliable datasets and building ingestion, transformation, and enrichment pipelines for analytics and AI workloads. Built production-grade systems integrating Snowflake, cloud services, and data platforms while ensuring data quality, validation, and operational reliability. Strong experience with LLM evaluation, monitoring, and observability using tools like LangSmith, Langfuse, Arize, TruLens, and Datadog, along with solid cloud and DevOps foundations (AWS, Azure, GCP, Docker, Kubernetes, CI/CD) focused on traceability, performance, and cost-efficient AI-driven applications.

PROFESSIONAL EXPERIENCE

03/2022- Present

Senior Full-Stack Developer | AI Engineer Freelancer - Remote

- Delivered production data and backend systems using Python and SQL, designing scalable APIs, data services, and processing layers supporting 10K+ requests/day with high availability and reliability.
- Designed and implemented data ingestion and processing pipelines integrating APIs, relational databases, and distributed data sources, enabling transformation of unstructured and semi-structured data into structured datasets for analytical and AI workloads.
- Built scalable data pipelines using Airflow and Databricks/PySpark (batch and streaming), integrating sources such as public APIs, SAP systems, relational databases, and data lakes while reducing refresh cycles by up to 80%.
- Implemented automated validation and reliability mechanisms (data quality checks, regression tests, monitoring) ensuring correctness of data outputs and improving operational stability across production data systems.
- Optimized data access layers across PostgreSQL, SQL Server, MongoDB, Redis, and Elasticsearch using indexing, caching, and query tuning, improving API and analytical latency by up to 75% under load.
- Architected and deployed cloud-native data platforms across Azure, AWS, and GCP with Docker, Kubernetes, and Terraform, enabling scalable distributed systems and reducing infrastructure costs by approximately 35%.
- Developed reusable data services and internal tools supporting large-scale ingestion, transformation, and reporting workflows, improving maintainability and enabling consistent data pipelines across projects.
- Implemented monitoring, evaluation, and observability practices for data and AI-driven systems, including validation of outputs, performance tracking, and reliability checks using tools such as LangSmith and Datadog.

- Led end-to-end technical design for data platforms and backend systems (data flows, processing layers, API contracts, and deployments), partnering with stakeholders to deliver scalable, production-grade data and AI solutions.

Tech keywords (ATS): **Python, SQL, Snowflake, FastAPI, TypeScript, Node.js, PySpark, Apache Spark, Airflow, Databricks, Kafka, Delta Lake, Data Pipelines, Data Processing, Data Validation, Data Quality, Docker, Kubernetes, Terraform, Azure, AWS, GCP, CI/CD, Observability, LangSmith, Datadog**

01/2026- Present

Senior Data Engineer

CI&T - Remote

- Built enterprise data platform foundations by codifying infrastructure and governance controls using Pulumi (TypeScript) across multiple environments, enabling reliable and repeatable deployments for cloud-based data and AI-ready systems.
- Implemented privacy-by-design controls in Snowflake (tag-based PII masking policies with deterministic hashing for string and numeric PII), ensuring secure downstream data consumption and supporting regulated enterprise environments.
- Designed and maintained secure data access patterns (RBAC, grants, environment separation, controlled privileged operations), enabling compliant and scalable access to operational and analytical datasets.
- Standardized secrets management and secure configuration (encrypted IaC configuration aligned with Azure Key Vault), reducing exposure risks and ensuring consistent deployments across distributed data platform components.
- Delivered end-to-end data ingestion pipelines with CDC via Fivetran to ADLS (Iceberg) and Snowflake (external volumes, catalog integration), enabling reliable ingestion and structuring of large-scale operational and semi-structured data.
- Improved platform reliability through CI/CD hardening (reusable GitHub Actions workflows, preview and refresh steps, environment gates, connector-specific credentials), enabling safer and repeatable infrastructure changes.
- Performed root-cause analysis and remediation of production incidents (privilege bootstrap gaps, deployment failures, environment drift), introducing preventative controls to increase platform stability and data reliability.
- Applied engineering automation and validation practices to support data governance and quality, ensuring infrastructure and data flows remained aligned with security, auditability, and operational standards.
- Maintained structured technical documentation and platform context across repositories, consolidating design decisions, operational procedures, and standards to improve team alignment and operational continuity.
- Enabled downstream AI and analytics use cases by structuring semi-structured and operational data within Snowflake, supporting reliable data consumption patterns and improving traceability, consistency, and usability of data across enterprise workloads.

Tech Keywords (ATS): **Data Platforms, Distributed Data Systems, Snowflake, Data Governance, RBAC, Compliance, Auditability, Pulumi, Infrastructure as Code, GitHub Actions, CI/CD, Azure Key Vault, Secrets Management, Fivetran, CDC, ADLS Gen2, Iceberg, Data Pipelines, Data Structuring, Data Reliability, Cloud Data Platforms**

09/2023- 01/2026

Senior Data Engineer (Full-Stack & Data Products)

Globant - Remote

- Developed and operated large-scale data ingestion and transformation platforms using Python, Azure Databricks, PySpark, Delta Lake, and SQL, processing 10M+ records per month from APIs, web sources, and enterprise systems with validation, deduplication, and schema governance.
- Designed and optimized distributed ETL and ELT pipelines orchestrated with Airflow and Azure Data Factory for structured and semi-structured data, standardizing ingestion workflows and reducing manual integration effort by approximately 80 percent.
- Built automated data validation, monitoring, and alerting frameworks including completeness checks, anomaly detection, schema drift detection, and SLA tracking, improving reliability across frequently changing data sources.
- Optimized SQL transformation layers, indexing strategies, and query performance for high-volume workloads, reducing execution time from approximately 18 seconds to under 3 seconds.
- Developed backend services and automation using Python and Azure Functions to orchestrate ingestion pipelines, manage retries, and handle failure recovery, improving resilience of distributed data systems.
- Implemented observability dashboards using Power BI and Elasticsearch with Kibana to monitor data freshness, pipeline health, error rates, and latency, reducing troubleshooting cycles from days to minutes.
- Collaborated with engineering, product, and operations teams in agile delivery, owning ingestion components end-to-end and contributing to architecture decisions for scalable data platforms.
- Applied software engineering best practices including version control, CI/CD, environment parity, and defensive programming to ensure reliable production-grade data systems.
- Implemented incremental ingestion strategies such as idempotent loads, checkpointing, and reprocessing, along with standardized dataset versioning, improving data recoverability and reducing operational overhead.
- Established data modeling and delivery standards for curated layers, aligning schema evolution and downstream consumption with analytics stakeholders while improving data consistency and reuse.

Tech keywords (ATS): Python, PySpark, Apache Spark, Azure Databricks, Delta Lake, SQL, Airflow, Azure Data Factory, Distributed Systems, Data Pipelines, Data Validation, Data Quality, Schema Governance, Schema Drift, Deduplication, Anomaly Detection, SLAs, Azure Functions, Observability, Power BI, Elasticsearch, Kibana, CI/CD

08/2021- 09/2023

Data Engineer

Santander Group - São Paulo

- Designed and delivered interactive Power BI dashboards and automated reporting pipelines using Python, reducing manual reporting effort by approximately 70 percent and expanding self-service analytics across finance and operations.
- Integrated and standardized data from SAP, external APIs, SQL databases, and Azure Data Lake, consolidating large-scale datasets into a centralized analytical environment with consistent definitions, governance, and traceability.

- Built distributed data processing pipelines using PySpark and SparkSQL, achieving three to five times performance improvements over legacy processes and accelerating insights for business stakeholders.
- Implemented scalable Azure-based data architectures using Synapse Analytics, ADLS, and Databricks, enabling near real-time data availability for financial and operational reporting.
- Developed automated data validation and anomaly detection routines, increasing trust in datasets to over 95 percent accuracy and reducing recurring data quality issues.
- Created and maintained KPI dashboards for finance, sales, and operations, shortening decision cycles from days to hours and improving alignment across teams.
- Collaborated with cross-functional stakeholders to define reporting requirements, maintain data dictionaries, document pipelines, and translate technical outputs into actionable insights.
- Implemented data quality controls including completeness, consistency, and reconciliation checks with monitoring for critical datasets, improving reliability and reducing reporting discrepancies.
- Optimized transformation logic and dataset refresh strategies such as incremental loads and partitioning, improving performance and supporting stable delivery of time-sensitive dashboards.

Tech keywords (ATS): Python, Power BI, PySpark, Apache Spark, SparkSQL, Azure, Databricks, Synapse Analytics, ADLS, SQL, SAP, ETL, Data Pipelines, Distributed Data Processing, Data Validation, Anomaly Detection, Data Quality, Data Governance, KPI Dashboards

03/2020- 02/2021

Data Analyst Intern

Novelis - São Paulo

- Consolidated data from Oracle, SQL/NoSQL systems, SAP, and shop-floor IoT sources, contributing to a near real-time analytical environment used to monitor operational performance across production lines.
- Developed automated Python scripts for data ingestion, standardization, and cleansing, reducing manual spreadsheet work by ~60% and accelerating updates for production and quality KPIs.
- Designed and delivered interactive Power BI dashboards for real-time tracking of production efficiency, machine downtime, and critical operational indicators, shortening analysis cycles from hours to minutes.
- Implemented automated data validation, reconciliation, and anomaly detection rules, increasing data accuracy and reliability to 95%+ and improving trust in operational reporting.
- Partnered with engineers and plant supervisors to define reporting requirements, document data sources, and maintain KPI dictionaries, translating raw operational data into actionable insights for non-technical teams.
- Implemented repeatable data processing workflows and standardized transformation logic for operational datasets, improving consistency, maintainability, and reliability of reporting pipelines.

TECHNICAL SKILLS

AI / LLM Engineering

- LLM Agents, Agent Orchestration (LangGraph), LangChain, Tool/Function Calling, Structured Outputs (JSON schema), Multi-step Workflows, Human-in-the-loop
- RAG: ingestion/parsing, chunking strategies, embeddings, vector DBs (Pinecone, Chroma), retrieval + reranking, grounding, citation-ready outputs
- Prompt Engineering, Evaluation & Guardrails (quality, hallucination/grounding, latency, cost), Observability (LangSmith, Langfuse, Arize, TruLens, Datadog)

Backend & APIs

- Python (FastAPI, Flask), Node.js (Express), REST APIs, GraphQL
- Authentication/Authorization: JWT, OAuth2, RBAC, Rate Limiting

Cloud & DevOps

- AWS, Azure, GCP, Docker, Kubernetes, Terraform
- CI/CD, GitHub Actions, Serverless (Azure Functions, AWS Lambda)
- Monitoring & Logging, Observability (Elasticsearch, Kibana, Datadog)

Data, Streaming & Storage

- SQL, Snowflake, PostgreSQL, SQL Server, MongoDB, Redis, Elasticsearch
- Apache Kafka, PySpark, SparkSQL, Delta Lake

Orchestration & Pipelines

- Apache Airflow, Azure Data Factory (ADF)
- Batch & Streaming Pipelines, Data Validation, Schema Governance

Frontend

- TypeScript, React, Next.js, Tailwind CSS, HTML5, CSS3, SSR/SSG, WebSockets

Testing & Automation

- Pytest, Jest, Playwright, Selenium, Automated QA, Data Quality Checks

LANGUAGES

- **English** — C1 (Professional working proficiency)
- **Spanish** — B1 (Intermediate)
- **Portuguese** — Native

EDUCATION

B.Sc. in Chemical Engineering
University of São Paulo (USP)